Arunabha Majumder

Mechanical Design Engineer

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Interstellar Lab, Paris

Application: Senior Mechanical Design Engineer

Dear Interstellar Lab team,

I am applying for the Senior Mechanical Design Engineer role. I am a mechanical engineer doing my PhD in Mechatronics and Robotics at Aalborg University, with my thesis expected to be finished in March. Your BioPods caught my eye because they are the kind of product I like most: a closed system where mechanics, electronics and software all have to work together, and where small design choices decide if it works reliably or not.

For the last 3 years I have owned the mechanical design of a hybrid shoulder exoskeleton. I came up with the idea of a spring carrying most of the arm load so a small 6 Nm motor is enough, and took it from hand sketches to SolidWorks assemblies and drawings. I 3D printed and rebuilt the parts many times, keeping track of each revision, and integrated the motor, encoder and force sensors with the control and software people. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I am happiest in the lab with the hardware in front of me. I built custom test rigs, implemented an Arduino-based trigger to sync a 5-bar robot with Qualisys motion capture, and collected data from 12 participants to check the design on real people. In my Bachelor's capstone I did the CAD, machining and assembly of a rehab exoskeleton myself.

I want to be honest that I do not have 6 years in industry, and I have not used Onshape yet, though moving from SolidWorks should be quick. I have not worked on plant growth or space hardware either. What I bring is full ownership of a design from sketch to tested prototype, and the habit of explaining ideas clearly. I have a Diploma in Fine Arts, so a quick sketch in a design review comes naturally to me.

Thank you for considering my application. My portfolio is at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder